COMPLETE TRADING LOGIC

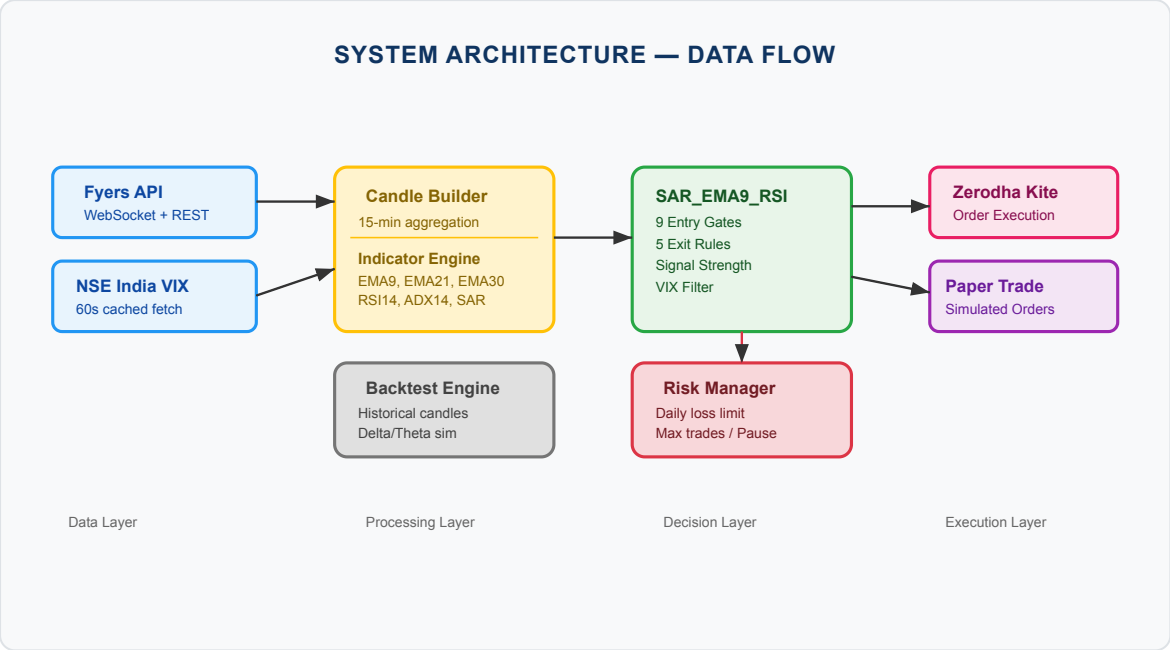
Palani Andawar Trading System — Full Conditions Reference

Strategy: SAR_EMA9_RSI | Instrument: NIFTY Options/Futures | v3.0 | March 2026

Table of Contents

1. System Architecture Overview
2. Trading Modes & Infrastructure
3. Indicators & Configuration
4. Pre-Trade Checks (Session Guards)
5. Entry Conditions — All 9 Gates
6. Signal Strength Classification
7. Stop Loss Initialization
8. Exit Conditions — Priority Ordered
9. Trailing Stop Loss — Tiered System
10. Risk Management Controls
11. VIX Market Regime Filter
12. Position Sizing & PnL Calculation
13. Time-Based Rules
14. Holiday & Expiry Handling
15. Edge Cases & Special Logic
16. Configuration Reference

1. System Architecture Overview



2. Trading Modes

Live Trading

Real-time Fyers WebSocket ticks. Orders executed via Zerodha Kite Connect API. Full risk controls active.

PRODUCTION

Paper Trading

Same logic as live. Simulated order fills at market price. No real money. Matches backtest behavior.

SIMULATION

Backtesting

Historical Fyers candles. Option premium simulation with configurable delta & theta decay.

HISTORICAL

3. Indicators & Configuration

Indicator	Period	Input	Purpose	Threshold
EMA9	9	OHLC4	Primary trigger (touch detection) + slope	Slope ≥ |6| pts
EMA21	21	OHLC4	Momentum alignment (EMA9 vs EMA21)	CE: EMA9 > EMA21
EMA30	30	Close	Medium-term trend filter	CE: Close > EMA30
RSI(14)	14	Close	Momentum confirmation	CE: >52 | PE: <48
ADX(14)	14	H/L/C	Trend strength filter	≥ 25
Parabolic SAR	—	H/L	Stop loss + trend direction	Step: 0.02, Max: 0.2
India VIX	—	NSE API	Market regime filter	Max: 20, Strong: 16

> **Why OHLC4 for EMAs?** OHLC4 = (Open+High+Low+Close)/4 is smoother than close-only EMA, reducing false touch signals during volatile candles. This produces fewer but higher-quality entries.

4. Pre-Trade Checks (Session Guards)

These checks run BEFORE any strategy evaluation begins each day.

SESSION START — PRE-TRADE VALIDATION FLOW

Holiday data: NSE API (https://www.nseindia.com/api/holiday-master?type=trading) with 24h cache + hardcoded 2026 fallback

5. Entry Conditions — All 9 Gates

Every potential entry must pass ALL 9 gates sequentially. Any single failure rejects the signal.

COMPLETE ENTRY GATE PIPELINE

1	**EMA9 Touch**	CE: Candle High ≥ EMA9 | PE: Candle Low ≤ EMA9 (checked on every tick)	✓
2	**SAR Position**	CE: SAR below price (trend=1) | PE: SAR above (trend=-1) | Or: SAR flipped this candle | Or: Bull Override	✓
3	**EMA9 Slope**	CE: Slope ≥ +6 pts (rising) | PE: Slope ≤ -6 pts (falling) — filters flat EMAs	✓
4	**Trend Filter**	CE: Close > EMA30 AND EMA9 > EMA21 | PE: Close < EMA30 AND EMA9 < EMA21	✓
5	**Candle Body**	CE: Close > Open (bullish) AND body ≥ 10pts | PE: Close < Open AND body ≥ 10pts — no dojis	✓
6	**SAR Distance**	Min: 45 pts | Max: 200 pts — SL not too tight (noise) or too wide (excess risk)	✓
7	**ADX Strength**	ADX(14) ≥ 25 — market must be trending, not ranging/choppy	✓
8	**RSI Momentum**	CE: RSI(14) > 52 | PE: RSI(14) < 48 — confirms directional momentum	✓

9	**VIX Filter**	VIX ≤ 20 to enter | VIX 16-20: STRONG signals only | VIX > 20: ALL blocked	✓
✓	**ENTRY CONFIRMED**— Place CE/PE option order at ATM strike via Zerodha		🚀

5.1 SAR Position — Three Valid Logics

Logic	Condition	Applies To	SL Source
Logic 1 Already Positioned	SAR already on correct side before entry candle. CE: trend=1 (dots below) | PE: trend=-1 (dots above)	CE & PE	SAR value
Logic 2 Just Flipped	SAR flipped ON this candle. CE: trend changed -1 → +1 | PE: trend changed +1 → -1	CE & PE	SAR value
Logic 3 Bull Override	SAR still bullish (trend=1) BUT strong down momentum: EMA9 slope ≤ -6, Close < EMA9, RSI < 42, (Close-SAR) > 50pts	PE Only	**EMA9** (not SAR)

6. Signal Strength Classification

SIGNAL STRENGTH DETERMINES ENTRY TIMING

STRONG Signal
EMA9 slope ≥ 9 AND RSI > 58 (CE) or < 40 (PE)
→ **→ Enter INTRA-CANDLE**
On tick while bar forms. Fastest fills. Always on 5-min TF.

MARGINAL Signal
EMA9 slope 6-8, RSI passes but not strong level
→ **→ Wait for CANDLE CLOSE**
Safer entry. Blocked when VIX 16-20.

7. Stop Loss Initialization

Standard SL (Logic 1 & 2)

```
stopLoss = currentSAR.value
initialStopLoss = stopLoss
trailActivatePts = min(max(gap ×
0.25, 12), 40)

// Example: Entry 25000, SAR 24950
// gap = 50, trailActivate =
max(12.5, 12) = 12pts
```

Logic 3 Override SL (PE only)

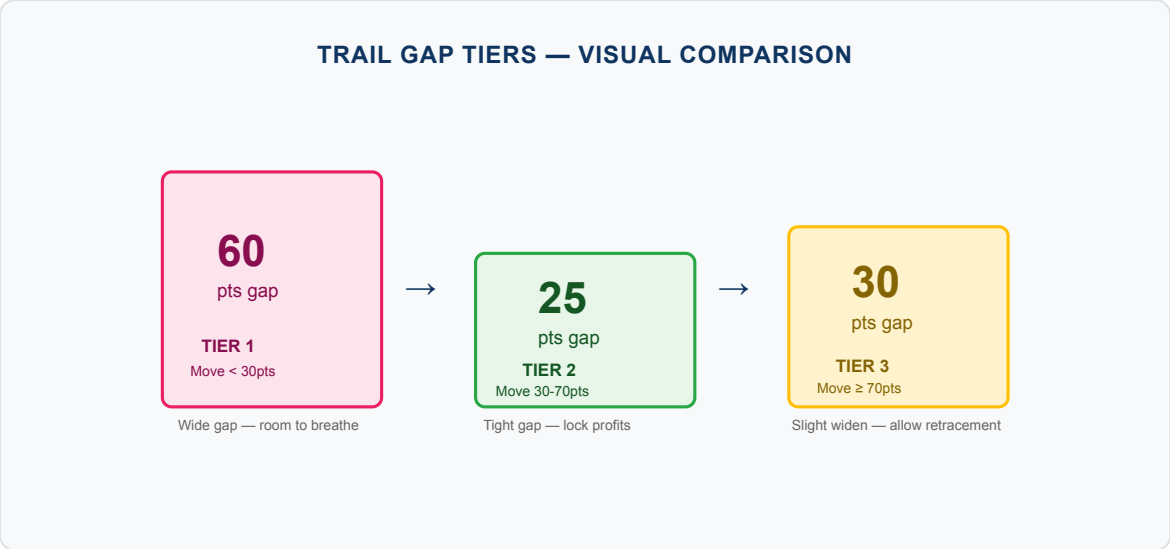
```
stopLoss = EMA9.value  // SAR too far
initialStopLoss = EMA9.value

// SAR is 50+ pts away from close
// EMA9 provides tighter, meaningful
SL
// Prevents oversized risk on PE
trades
```

8. Exit Conditions — Priority Ordered

Checked on **every tick**. First matching rule triggers the exit.

Priority	Exit Type	Condition	Exit Price
P1	**Breakeven Stop**	Move in favour ≥ 25pts → SL moves to entry price. Not an exit itself, but prevents loss.	—
P2	**Trailing SL**	Move ≥ trailActivatePts → SL = bestPrice ∓ dynamicGap. Gap tiers: T1=60, T2=25, T3=30. Only tightens.	Trail SL price
P3	**Stop Loss Hit**	CE: LTP ≤ stopLoss | PE: LTP ≥ stopLoss → Immediate exit	SL price
P4	**Opposite Signal**	CE position gets BUY_PE signal (or vice versa) → Exit at candle close	Candle close
P5	**EOD Square-Off**	Time ≥ 3:20 PM IST → Force close all positions	Market price

9. Trailing Stop Loss — Tiered Dynamic Gap System




```
// Trail SL calculation (CE example):
trailSL = bestPrice - dynamicGap;
if (trailSL > currentSL)   // Can ONLY tighten
    currentSL = trailSL;

// Trail activation:
trailActivatePts = min(max(SARgap * 0.25, 12), 40);
// 50pt SAR gap  → activates at 12pts of profit
// 160pt SAR gap → activates at 40pts of profit
```

9.1 Trail Activation — Dynamic Per Trade

SAR Gap	Calculation	Trail Activates At
45 pts	max(45×0.25, 12) = max(11.25, 12)	**12 pts**
80 pts	max(80×0.25, 12) = max(20, 12)	**20 pts**
120 pts	max(120×0.25, 12) = max(30, 12)	**30 pts**
200 pts	min(max(200×0.25, 12), 40) = min(50, 40)	**40 pts** (capped)

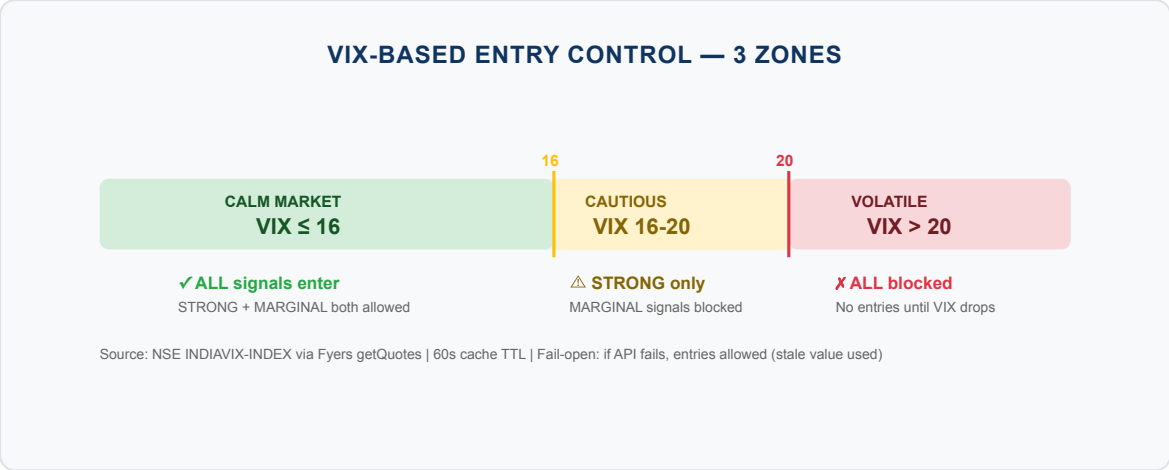
10. Risk Management Controls

Daily Loss Limit
₹5,000
Hard stop for entire day
Latches ON, resets next day
sessionPnl ≤ -MAX_DAILY_LOSS
→ Block ALL new entries
Env: MAX_DAILY_LOSS

Max Daily Trades
20
Hard cap per session
Prevents overtrading
sessionTrades.length ≥ 20
→ Block ALL new entries
Env: MAX_DAILY_TRADES

Consecutive Loss Pause
3+
losses in a row → Pause
15-minute cooling period
consecutiveLosses ≥ 3
→ Pause entries 15 min
Env: MAX_LOSS_PAUSE_MS

11. VIX Market Regime Filter



Why VIX Filter? High VIX = choppy market where price whipsaws through EMA9, SAR flips every 1-2 candles, and trailing SL gets hit prematurely. Result: string of small losses. Better to sit out entirely or only take highest-conviction signals.

12. Position Sizing & PnL Calculation

Position Size

```
lotQty = NIFTY_LOT_SIZE ×
LOT_MULTIPLIER
       = 65 × 1 = 65 shares/contract

// Configurable via:
// NIFTY_LOT_SIZE (default: 65)
// LOT_MULTIPLIER (default: 1)
```

Brokerage

```
BROKERAGE = ₹80 per trade
(entry+exit)

// Applied in backtest/PnL calc
// Realistic estimate for Zerodha
// options turnover
```

PnL Calculation — Three Modes

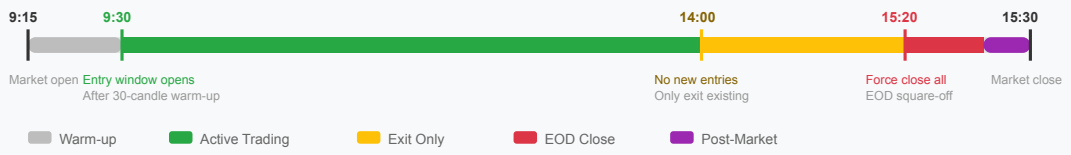
Mode	Formula	When Used
Futures	`PnL = (exitPrice − entryPrice) × direction × LOT_SIZE − ₹80` direction: CE=+1, PE=-1	INSTRUMENT = NIFTY_FUTURES

Options (Sim)	`spotPnl = (exit − entry) × direction`
	`premiumMove = spotPnl × DELTA (0.55)`
	`thetaDecay = (THETA_DAY/26) × candlesHeld`
	`PnL = (premiumMove − thetaDecay) × LOT_SIZE − ₹80`
		BACKTEST_OPTION_SIM = true

Raw Points	`PnL = spotPnlPts` (no delta/theta/lot)	BACKTEST_OPTION_SIM = false

Option Sim defaults: Delta = 0.55 (ITM approximation), Theta = ₹10/day, Candles/day = 26 (15-min over 6.5 hours). Without simulation, a 100pt NIFTY move shows 100pt profit, but real option only gains ~55pt × ₹65 = ₹3,575.

13. Time-Based Rules



Rule	Time (IST)	Action	Reason
Market Open	9:15 AM	Start candle collection	NSE session begins
Entry Start	9:30 AM	Begin signal evaluation	30 candles needed for ADX warm-up
Entry Stop	2:00 PM	No new entries	Late entries don't have time for trail to work
EOD Close	3:20 PM	Force close all positions	Avoid overnight/expiry risk
Market Close	3:30 PM	Session ends	NSE session ends

14. Holiday & Expiry Handling

Holiday Detection

- **Primary:** NSE API `nseindia.com/api/holiday-master?type=trading`
- **Cache:** 24 hours
- **Fallback:** Hardcoded 2026 holiday list
- Checked at session start via `isTradingAllowed()`
- Weekend check: Saturday & Sunday blocked

Expiry Handling

- **NIFTY weekly expiry:** Auto-detected
- `getNearestThursdayExpiry()` computes next expiry
- Rolls to next expiry after 3:30 PM on expiry day
- Falls back if option chain unavailable
- ATM strike selected at entry time

15. Edge Cases & Special Logic

Edge Case	Logic	Why
Intra-Candle Throttle (15-min only)	Re-run signal eval only when bar high/low changes (not every tick). On 5-min TF, all signals enter intra-candle.	Prevents premature entries on partial candle data. Keeps tick processing fast.

SL Re-entry Block	If SL hit on a candle with NO trail activation (pure loss), block re-entry on SAME candle.	Prevents revenge trading on noise. If trail was active (partial profit), re-entry IS allowed.
Option Symbol Pre-fetch	After each candle close, async pre-fetch CE & PE symbols for next potential entry. Force refresh if spot moves >25pts from cached value.	Eliminates REST API delay at entry time. Improves fill speed.
Signal Missed Logging	MARGINAL signals that can't enter intra-candle are logged (once per candle) for diagnostics.	Helps diagnose missed opportunities without spam.
VIX Fail-Open	If VIX API fetch fails, use stale cached value. If no cache exists, allow entry.	Prefer trading with imperfect data over missing moves due to API hiccup.
Trail Only Tightens	Trail SL can only move in the protective direction. Never loosens even if dynamic gap changes tiers.	Prevents giving back locked-in profit on tier boundary transitions.

16. Complete Configuration Reference

Trading Parameters

Env Variable	Default	Description
TRADE_RESOLUTION	15	Candle timeframe in minutes (5 or 15)
TRADE_START_TIME	09:15	Market open time (HH:MM)
TRADE_STOP_TIME	15:30	Market close time (HH:MM)
LIVE_TRADE_ENABLED	false	Enable live order execution
INSTRUMENT	NIFTY_OPTIONS	NIFTY_OPTIONS or NIFTY_FUTURES
NIFTY_LOT_SIZE	65	Lot size per contract
LOT_MULTIPLIER	1	Number of lots per trade

Strategy (SAR_EMA9_RSI)

Env Variable	Default	Description
MAX_SAR_DISTANCE	200	Max allowed SAR-to-entry gap (points)
BREAKEVEN_PTS	25	Points of profit before SL moves to entry

Trailing Stop Configuration

Env Variable	Default	Description
TRAIL_TIER1_UPTO	30	Move threshold: Tier 1 → Tier 2 (points)
TRAIL_TIER2_UPTO	70	Move threshold: Tier 2 → Tier 3 (points)
TRAIL_TIER1_GAP	60	Trailing gap for Tier 1 (points)
TRAIL_TIER2_GAP	25	Trailing gap for Tier 2 (points)
TRAIL_TIER3_GAP	30	Trailing gap for Tier 3 (points)
TRAIL_ACTIVATE_PTS	12	Minimum move for trail activation (points)

Risk Management

Env Variable	Default	Description
MAX_DAILY_LOSS	5000	Daily loss limit in ₹ — blocks all entries when hit
MAX_DAILY_TRADES	20	Maximum trades per day
MAX_LOSS_PAUSE_MS	900000	Pause duration after 3+ consecutive losses (15 min)

VIX Filter

Env Variable	Default	Description
VIX_FILTER_ENABLED	true	Enable/disable VIX-based entry filtering
VIX_MAX_ENTRY	20	Block ALL entries when VIX exceeds this
VIX_STRONG_ONLY	16	Only STRONG signals when VIX exceeds this

Backtest

Env Variable	Default	Description
BACKTEST_OPTION_SIM	true	Enable delta/theta option simulation
BACKTEST_DELTA	0.55	Option delta approximation (ITM)
BACKTEST_THETA_DAY	10	Daily theta decay in ₹

Palani Andawar Trading System — Complete Trading Logic Documentation — Generated March 2026
All parameters configurable via environment variables (.env file)